

AI/ML Decision Tree Classifier (CLI)

Project Overview

This project is a fully command-line executable Machine Learning tool that uses a Decision Tree Classifier to predict categorical outcomes based on numerical inputs. It utilizes the classic Iris botanical dataset, trained dynamically at runtime, to ensure there are no external dependencies on local CSV files.

Environment Setup & Prerequisites

To run this project, Python 3.8 or higher is required.

1. Create a Virtual Environment (Recommended)

Open your terminal, navigate to the root directory of this repository, and run:

```
python -m venv venv
```

Activate the virtual environment:

- **Windows:** `venv\Scripts\activate`
- **macOS/Linux:** `source venv/bin/activate`

2. Install Dependencies

Run the following command to install the required Machine Learning library:

```
pip install scikit-learn numpy
```

Execution Instructions

This project is built strictly for the terminal. There is no GUI. It requires four numerical arguments representing physical measurements (in cm).

To evaluate the model, run `main.py` and provide the required arguments: `--sepal-length`, `--sepal-width`, `--petal-length`, and `--petal-width`.

Example 1: Predicting Class 1 (Setosa)

```
python main.py --sepal-length 5.1 --sepal-width 3.5 --petal-length 1.4 --petal-width 0.2
```

Example 2: Predicting Class 3 (Virginica)

```
python main.py --sepal-length 6.5 --sepal-width 3.0 --petal-length 5.2 --petal-width 2.0
```

Expected Output Format

The terminal will display the input array, the predicted classification, and the model's confidence percentage:

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DECISION TREE CLASSIFIER RESULT

Input Features : [5.1, 3.5, 1.4, 0.2]

**Predicted Class: SETOSA Confidence :
100.00%**